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A Structural Timeline of the Universe

Sequential Structural Explosions in Cosmic Evolution

Abstract

The history of the universe contains several episodes of rapid diversification and complexity growth. These include the emergence of matter following the Big Bang, the formation of chemical complexity in stars, the emergence of life, the rapid diversification of animal body plans during the Cambrian Explosion, and the technological transformation associated with the Industrial Revolution.

Within the Structural Limits framework, these events can be interpreted as **structural explosions**: transitions between domains in the global state space of complex systems. Each transition dramatically expands the set of accessible configurations. This paper proposes a structural timeline of the universe where major historical transitions correspond to expansions of structural possibility space.

1. Introduction

The universe does not evolve smoothly.

Instead, its history appears punctuated by **bursts of novelty**:

- new physical structures
- new chemical possibilities
- new biological forms
- new cognitive systems.

These bursts suggest that cosmic evolution may proceed through **transitions between structural regimes** rather than gradual change alone.

2. Structural Domains in Cosmic Evolution

At any moment in cosmic history, the universe occupies a region of its possible configuration space.

Each region corresponds to a **structural regime** where certain organizations are possible and others are not.

Transitions between regimes occur when the system reaches **structural limits**, allowing access to a larger space of configurations.

3. The Structural Timeline

A simplified structural timeline of the universe may look like this:

```
cosmic origin
  ↓
physical structure explosion
  ↓
chemical complexity explosion
  ↓
biological origin
  ↓
morphological explosion (Cambrian)
  ↓
cognitive explosion
  ↓
technological explosion
```

Each stage corresponds to an **expansion of structural possibilities**.

4. Cosmological Explosion

The earliest transition occurs after the Big Bang, when fundamental particles, atoms, and later stars and galaxies emerge.

This stage opens the structural domain of **physical matter organization**.

5. Chemical Explosion

Inside stars, nuclear processes generate heavier elements.

This enables rich chemical networks, dramatically expanding the number of possible molecular structures.

The universe enters the domain of **complex chemistry**.

6. Biological Explosion

The emergence of life represents another expansion in configuration space: self-replicating molecular systems capable of evolution.

Later, the Cambrian Explosion rapidly expands the morphological possibilities of multicellular organisms.

7. Cognitive Explosion

The emergence of symbolic cognition and language in humans expands the space of behavioral and informational possibilities.

Culture becomes a new structural domain.

8. Technological Explosion

Beginning with the Industrial Revolution, technological systems evolve rapidly, generating new artificial structures and accelerating complexity growth.

9. Structural Pattern

Across all these events, the same sequence appears:

```
stable regime  
→ structural instability  
→ explosive diversification  
→ new structural regime
```

This pattern suggests that cosmic evolution may be governed by **structural transitions between domains of possibility**.

10. Implications

Viewing the universe through structural explosions suggests:

- complexity grows in **punctuated bursts**
 - apparent randomness reflects exploration near structural limits
 - major evolutionary transitions correspond to expansions of possibility space.
-

11. Conclusion

The history of the universe can be interpreted as a sequence of structural regime transitions.

Each transition dramatically expands the accessible configuration space, producing the explosive emergence of new forms of complexity.

Understanding these transitions may provide a unified framework linking cosmology, biology, and technological evolution.

Figure 1 — Structural Timeline of the Universe

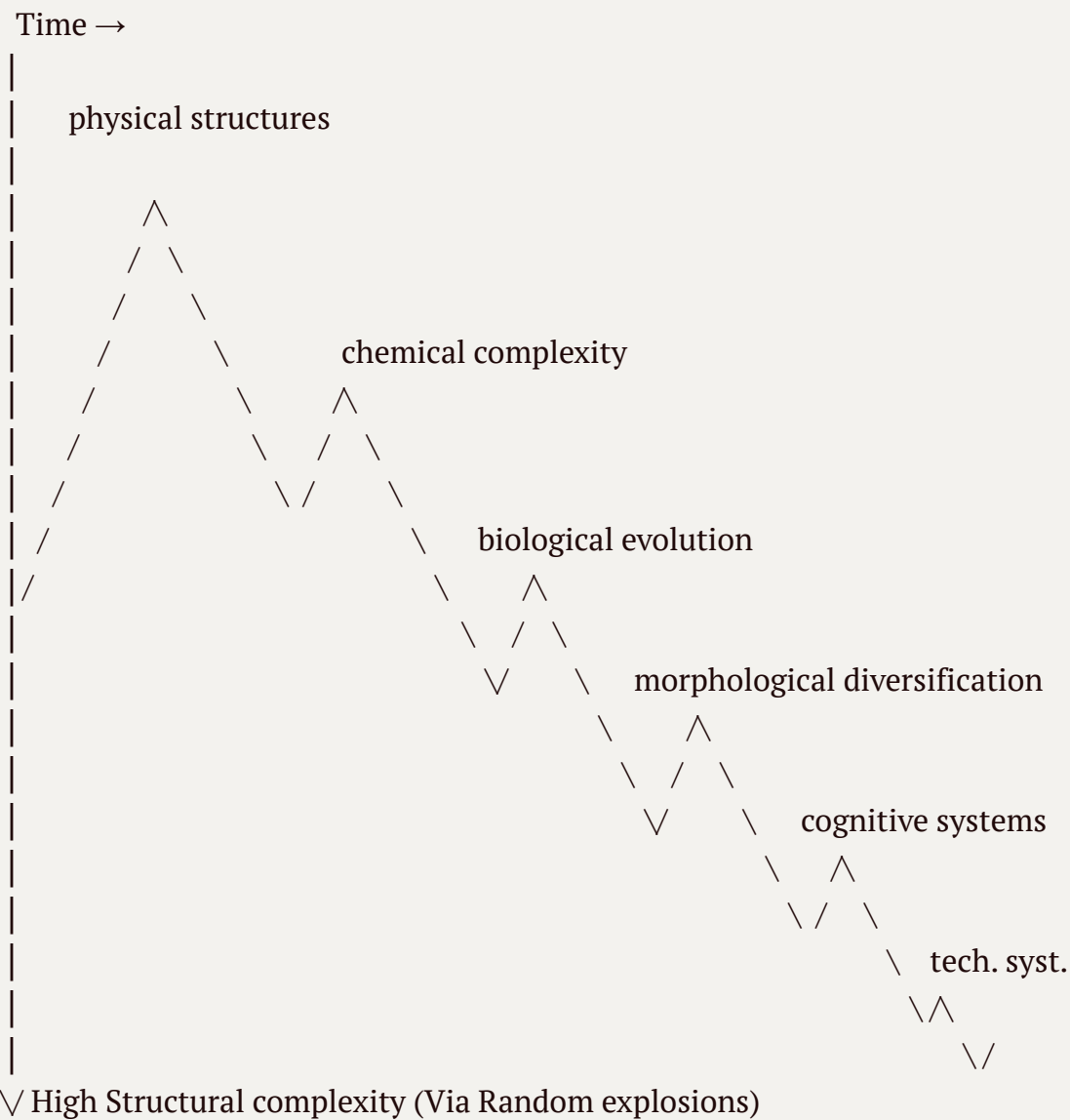
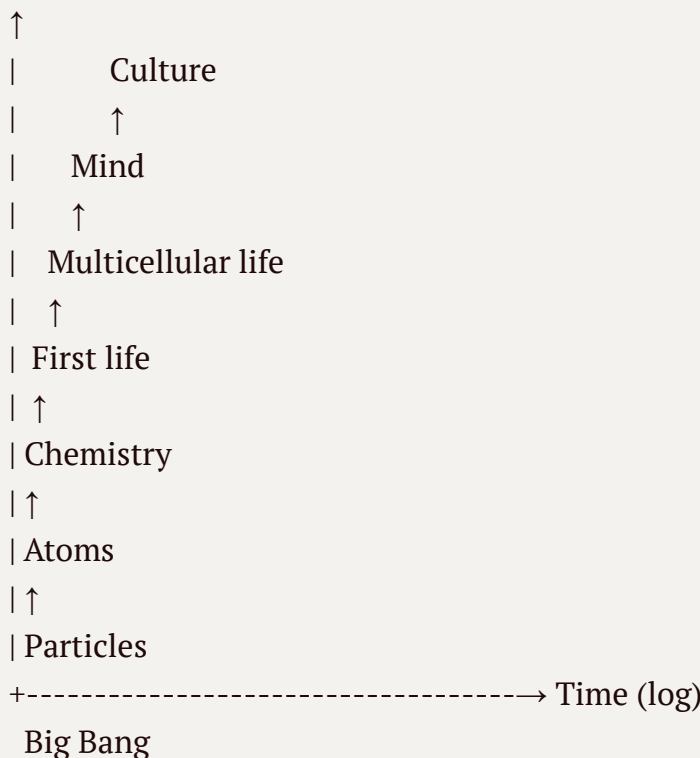


Figure 1 | Structural timeline of the universe.

The history of the universe can be interpreted as a sequence of structural expansions. Each explosion corresponds to a transition between structural domains in the global configuration space. These transitions dramatically increase the number of accessible configurations, producing rapid diversification across physical, chemical, biological, cognitive, and technological systems.

Structural Complexity



Empirical Law of Structural Acceleration

When all known structural explosions of the universe are placed on a diagram of **cosmological time** x versus **structural complexity** y , a striking empirical regularity appears:

the intervals between structural explosions decrease approximately in a logarithmic or inverse-exponential manner.

In other words,

■ *each new structural level tends to emerge faster than the previous one.*

This produces a **temporal compression of structural transitions** throughout cosmic history.

Approximate Sequence of Structural Explosions

EVENT	TIME SINCE BIG BANG
Big Bang / elementary particles	10^{-43} s
Formation of protons and neutrons	10^{-6} s
Atomic recombination	3.8×10^5 years
First stars	10^8 years
Complex chemistry	10^9 years
Origin of life	10^{10} years
Cambrian explosion	1.35×10^{10} years
Human intelligence	1.38×10^{10} years
Technological civilization	nearly instantaneous on cosmological scales

If the **intervals between events** are estimated, they follow an approximately contracting sequence:

$$10^8 \text{ years}, 10^9 \text{ years}, 10^8 \text{ years}, 10^7 \text{ years}, 10^5 \text{ years}, 10^3 \text{ years}, \text{ decades}$$

The temporal scale **contracts rapidly** as structural complexity increases.

Empirical Regularity

This empirical pattern can be approximated by

$$\Delta t_n \propto e^{-kn}$$

where

- n denotes the structural level,
- Δt_n is the interval until the next structural explosion.

Equivalently,

$$t_n \sim \log(n)$$

when time is measured from the origin of the universe.

Structural Interpretation

Within the structural framework, this behavior emerges naturally from the growth of the **structural state space** Ω_s .

Each new domain increases the number of accessible configurations:

$$|\Omega_s^{n+1}| \gg |\Omega_s^n|$$

As a consequence, the probability of discovering a new structural domain increases:

$$P(\text{new domain}) \uparrow$$

because the system can explore a vastly larger configuration space.

The result is an **acceleration of structural innovations**.

Dynamical Consequence

The universe therefore does not evolve linearly in structural terms.

Instead, it follows a cascade of **accelerated structural transitions**, reminiscent of phenomena known in complex systems as:

- innovation cascades
- critical transitions
- structural avalanches

However, in the present framework these events correspond specifically to encounters with the **boundary of the structural domain**

$$\partial\Omega_s.$$

Crossing such boundaries allows the emergence of new structural regimes.

Empirical Law of Structural Acceleration

Let t_n denote the time at which structural level n appears.

Empirically,

$$t_{n+1} - t_n \approx Ae^{-kn}.$$

This implies

$$\lim_{n \rightarrow \infty} (t_{n+1} - t_n) = 0.$$

In other words,

structural transitions tend to concentrate in progressively shorter temporal scales.

Philosophical Implication

This result suggests that the universe may approach a regime in which

$$\frac{dC}{dt} \rightarrow \infty,$$

where C denotes structural complexity.

This resembles ideas such as the **technological singularity**, but in this framework it emerges naturally from the **expansion of structural state space**.

Structural Interpretation of Major Evolutionary Events

From this perspective, the **Cambrian explosion** is not an isolated anomaly but one instance in a broader series of structural transitions, including:

- nucleosynthesis
- complex chemistry
- origin of life
- Cambrian explosion
- symbolic cognition
- culture
- technology

All of these events correspond to **transitions between structural domains** within the evolving state space of the universe.

Structural Expansion and Complexity Acceleration

A key implication of the structural framework is that the acceleration of complexity is not accidental but emerges naturally from the **expansion of the structural state space** Ω_s .

Let

$$C(t)$$

denote the structural complexity of the universe at time t , and

$$|\Omega_s(t)|$$

the volume of accessible structural configurations.

We assume that complexity grows with the number of accessible configurations, so that

$$C(t) \sim \log |\Omega_s(t)|.$$

This logarithmic dependence reflects the fact that complexity measures the **information required to specify a structural configuration** within the domain.

Structural Expansion

When a system approaches the boundary of its structural domain

$$\partial\Omega_s$$

the **Structural Collapse Theorem** implies that the previous configuration loses dynamical stability.

At that point, two possibilities arise:

1. structural collapse
2. structural expansion into a higher-dimensional domain

In the second case,

$$|\Omega_s^{n+1}| \gg |\Omega_s^n|.$$

This expansion dramatically increases the number of accessible configurations.

Complexity Growth Law

Substituting into the complexity relation gives

$$C_{n+1} - C_n \sim \log \left(\frac{|\Omega_s^{n+1}|}{|\Omega_s^n|} \right).$$

If the structural domain grows multiplicatively,

$$|\Omega_s^{n+1}| = \alpha |\Omega_s^n|$$

with $\alpha \gg 1$, then

$$\Delta C_n \sim \log \alpha.$$

However, because the **exploration rate of configuration space also increases**, the temporal spacing between structural transitions decreases.

Acceleration of Structural Complexity

Let

$$\tau_n$$

denote the time required for the system to explore enough of Ω_s^n to reach the next boundary.

Assuming exploration scales inversely with available configurations,

$$\tau_n \propto \frac{1}{|\Omega_s^n|}.$$

Therefore

$$\tau_{n+1} \ll \tau_n.$$

This yields

$$\frac{dC}{dt} \uparrow$$

over cosmic time.

In the limit,

$$\frac{dC}{dt} \rightarrow \infty.$$

Connection with the Structural Collapse Theorem

The **Structural Collapse Theorem** states that when a system reaches the boundary of its structural domain, its governing dynamics lose stability or uniqueness.

Formally,

$$x(t) \rightarrow \partial\Omega_s$$

implies loss of stable continuation within the same structural regime.

This loss of stability forces the system into either:

- collapse (loss of structure), or
- expansion into a new domain Ω_s^{n+1} .

In the expansion case, the newly accessible configuration space increases dramatically, producing both

1. **structural innovation**, and
 2. **accelerated exploration of configuration space**.
-

Result: Structural Acceleration Principle

Combining these results yields a general principle:

Structural Acceleration Principle

$$\frac{dC}{dt} \propto \frac{d}{dt} \log |\Omega_s(t)|$$

which implies that **the growth of structural complexity is driven by the expansion of accessible structural state space**.

Structural explosions therefore occur when the system reaches

$$\partial\Omega_s$$

and transitions into a domain with dramatically larger configuration volume.

Consequence for the Structural Timeline

Within this framework:

- nucleosynthesis
- chemistry
- life
- multicellular organisms
- cognition
- culture
- technology

are all interpreted as **successive expansions of Ω_s** .

Each expansion increases the dimensionality of the structural space and accelerates the rate of subsequent innovations.

Master Equation of Structural Evolution

Let

$$\Omega_s(t)$$

denote the accessible **structural state space** of the universe at time t , and

$$C(t)$$

the structural complexity of the system.

Assume complexity scales with the logarithm of the number of accessible configurations:

$$C(t) = \log |\Omega_s(t)|.$$

Taking the time derivative gives

$$\frac{dC}{dt} = \frac{1}{|\Omega_s|} \frac{d|\Omega_s|}{dt}.$$

Structural Evolution Equation

We propose that the expansion of the structural domain is driven by encounters with its boundary:

$$\frac{d|\Omega_s|}{dt} = \Phi(\partial\Omega_s, t)$$

where

Φ represents the **rate at which new structural configurations become accessible when the system reaches structural limits**.

Substituting into the previous relation yields the **Master Equation of Structural Evolution**:

$$\boxed{\frac{dC}{dt} = \frac{\Phi(\partial\Omega_s, t)}{|\Omega_s|}}$$

Interpretation

This equation states that **the rate of complexity growth is governed by the interaction between the system and the boundary of its structural domain.**

Three regimes emerge naturally.

1 Stable structural regime

If the system remains far from the boundary,

$$\Phi(\partial\Omega_s, t) \approx 0$$

and

$$\frac{dC}{dt} \approx 0$$

Complexity grows slowly.

2 Structural explosion

When the system approaches the boundary,

$$x(t) \rightarrow \partial\Omega_s$$

the rate of new accessible configurations increases sharply:

$$\Phi \gg 0$$

leading to

$$\frac{dC}{dt} \uparrow$$

This corresponds to events such as

- origin of life
 - Cambrian explosion
 - emergence of intelligence.
-

3 Structural collapse

If the system cannot transition to a higher structural domain, the **Structural Collapse Theorem** applies.

The state leaves Ω_s :

$$x(t) \notin \Omega_s$$

which results in a rapid decrease in structural organization.

Structural Acceleration

Because structural transitions typically produce

$$|\Omega_s^{n+1}| \gg |\Omega_s^n|$$

the exploration rate of configuration space increases, producing

$$\frac{dC}{dt} \uparrow$$

over cosmic time.

In the limit,

$$\frac{dC}{dt} \rightarrow \infty$$

which corresponds to the **compression of structural transitions observed in the structural timeline of the universe**.

Compact Form

The structural evolution of the universe can therefore be summarized as

$$\boxed{\frac{d}{dt} \log |\Omega_s| = \frac{\Phi(\partial\Omega_s, t)}{|\Omega_s|}}$$

This relation expresses the central idea of the framework:

Complexity grows when systems interact with the structural limits of their accessible configuration space.

Structural Thermodynamics

Define the **structural entropy**

$$S_s = \log |\Omega_s|$$

where $|\Omega_s|$ is the number (or measure) of accessible structural configurations.

Then the evolution equation becomes

$$\frac{dS_s}{dt} = \frac{1}{|\Omega_s|} \frac{d|\Omega_s|}{dt}.$$

Substituting the structural expansion law

$$\frac{d|\Omega_s|}{dt} = \Phi(\partial\Omega_s, t)$$

gives

$$\frac{dS_s}{dt} = \frac{\Phi(\partial\Omega_s, t)}{|\Omega_s|}.$$

Structural First Law

The structural entropy changes only when the system interacts with the **boundary of its structural domain**.

Thus we can write a differential form:

$$dS_s = \frac{1}{|\Omega_s|} d|\Omega_s|$$

which mirrors the form of thermodynamic relations where entropy changes with the number of accessible microstates.

Structural Flux

Define the **structural innovation flux**

$$J_s = \Phi(\partial\Omega_s, t).$$

Then the fundamental equation becomes

$$\frac{dS_s}{dt} = \frac{J_s}{|\Omega_s|}.$$

This equation states that **structural entropy increases when innovation flows across structural boundaries**.

Structural Phase Transitions

When the system approaches the structural boundary

$$x(t) \rightarrow \partial\Omega_s$$

the flux J_s increases dramatically.

This produces a **structural phase transition**, corresponding to events such as

- origin of life
 - Cambrian explosion
 - emergence of cognition
 - technological civilization.
-

Structural Temperature

A useful derived quantity is the **structural temperature**

$$T_s = \left(\frac{\partial S_s}{\partial C} \right)^{-1}.$$

High structural temperature corresponds to regimes where small increases in complexity open large regions of structural state space.

These regimes are typically observed near structural explosions.

Structural Thermodynamic Identity

The full differential relation can therefore be written as

$$dS_s = \frac{J_s}{|\Omega_s|} dt.$$

This expression summarizes the central mechanism of structural evolution:

complexity grows through the expansion of structural configuration space when systems encounter structural limits.

Second Law of Structural Thermodynamics

Let

$$\Omega_s(t)$$

denote the accessible **structural state space** of the universe at time t , and define the **structural entropy**

$$S_s(t) = \log |\Omega_s(t)|.$$

Theorem (Second Law of Structural Evolution)

For any open dynamical system capable of structural innovation,

$$\boxed{\frac{dS_s}{dt} \geq 0}$$

that is,

$$\frac{d}{dt} \log |\Omega_s| \geq 0.$$

Interpretation

This inequality states that:

the volume of accessible structural configuration space tends to increase over time.

In other words, the universe tends to **expand the range of structures that can exist or be explored**.

Structural Meaning

The increase of structural entropy occurs when the system reaches the boundary of its current structural domain

$$x(t) \rightarrow \partial\Omega_s.$$

At this boundary, one of two events occurs:

1. **Structural collapse** (loss of accessible configurations), or
2. **Structural expansion** into a higher-dimensional domain.

Only the second case contributes to the long-term evolution of complexity.

Cosmological Interpretation

Over cosmic time, the universe repeatedly expands its structural state space through transitions such as:

- nucleosynthesis
- chemical complexity
- origin of life
- multicellular organisms
- cognition
- technological civilization.

Each transition increases $|\Omega_s|$, and therefore

$$S_s = \log |\Omega_s|$$

grows monotonically.

Relation to Complexity

If structural complexity C is defined as

$$C \sim S_s,$$

then the second law implies

$$\frac{dC}{dt} \geq 0.$$

Thus **the long-term tendency of the universe is toward increasing structural complexity**, driven by the expansion of accessible configuration space.

Compact Statement

The Second Law of Structural Thermodynamics can therefore be expressed as

$$\boxed{\frac{d}{dt} \log |\Omega_s| \geq 0}$$

which states:

The universe tends to expand its accessible structural state space.

Variational Principle of Structural Evolution

Let

$$S_s(t) = \log |\Omega_s(t)|$$

denote the **structural entropy**, where Ω_s is the accessible structural configuration space.

Define the **structural action functional**

$$\mathcal{A}_s = \int_{t_0}^{t_1} \frac{dS_s}{dt} dt.$$

Principle of Maximum Structural Expansion

The dynamical trajectory of the universe satisfies

$$\boxed{\delta \mathcal{A}_s = 0}$$

with the constraint

$$\frac{dS_s}{dt} \geq 0.$$

Interpretation

Among all possible dynamical trajectories in state space, the realized trajectory is the one that **extremizes the accumulated growth of structural entropy**.

In physical terms:

The universe evolves along paths that maximize the expansion of accessible structural configurations.

Equivalent Form

Since

$$\mathcal{A}_s = \int_{t_0}^{t_1} \frac{d}{dt} \log |\Omega_s(t)| dt$$

the principle can be written as

$$\delta \int_{t_0}^{t_1} \frac{d}{dt} \log |\Omega_s(t)| dt = 0.$$

Structural Interpretation

This principle implies that systems tend to evolve toward configurations that:

- increase the number of accessible structural states
- expand the dimensionality of structural space
- create new domains of organization.

When a system approaches the boundary

$$\partial\Omega_s$$

the structural dynamics may trigger:

- **structural collapse**, or
- **structural expansion**.

The latter corresponds to **structural explosions**, where the volume of accessible configuration space increases abruptly.

Cosmological Meaning

Under this variational principle, major transitions in the universe can be interpreted as **solutions maximizing structural entropy growth**, including:

- atomic structure formation
- chemical complexity
- origin of life

- biological diversification
- cognition and symbolic culture
- technological civilization.

Each transition corresponds to an **expansion of Ω_s** .

Compact Formulation

The principle of structural evolution can therefore be summarized as

$$\delta \int_{t_0}^{t_1} dS_s = 0$$

with

$$\frac{dS_s}{dt} \geq 0.$$

Physical Meaning

This principle suggests that **structural evolution is not arbitrary**.

Instead, it follows a directional tendency:

the progressive expansion of accessible structural possibilities.

Boundary Emergence of the Structural Collapse Theorem

1. Structural Variational Principle

Let the **structural action functional** be defined as

$$\mathcal{A}_s = \int_{t_0}^{t_1} \frac{dS_s}{dt} dt$$

where

$$S_s = \log |\Omega_s|$$

is the **structural entropy**, defined as the logarithm of the accessible structural state space Ω_s .

The evolution of the system is assumed to follow the variational principle

$$\delta \mathcal{A}_s = 0$$

subject to the structural second law

$$\frac{dS_s}{dt} \geq 0.$$

This principle implies that dynamical trajectories tend to **extremize the growth of accessible structural configurations**.

2. Structural Domains and Boundaries

Assume the system evolves within a structural domain

$$x(t) \in \Omega_s$$

with boundary

$$\partial \Omega_s.$$

While the trajectory remains inside the domain,

$$x(t) \in \text{int}(\Omega_s),$$

the system dynamics are well-defined and the governing equations admit smooth solutions.

However, when the trajectory approaches the structural boundary,

$$x(t) \rightarrow \partial\Omega_s,$$

the variation of the action functional acquires boundary contributions:

$$\delta\mathcal{A}_s = (\text{interior terms}) + (\text{boundary terms})$$

which cannot generally vanish within the original domain.

3. Critical Boundary Behavior

At the structural boundary, one of two generic outcomes occurs.

Case A — Loss of Dynamical Solutions

The governing equations cease to admit solutions within Ω_s .

This results in **structural collapse**, where the system can no longer persist in the existing configuration space.

Case B — Emergence of a New Structural Domain

A new configuration space becomes accessible:

$$\Omega_{s+1}$$

such that

$$|\Omega_{s+1}| \gg |\Omega_s|.$$

This produces a **structural explosion**, corresponding to the emergence of a higher-order structural regime.

4. Structural Collapse Theorem (Variational Form)

Theorem — Structural Collapse as Boundary Condition

Let a dynamical system evolve according to the structural variational principle

$$\delta\mathcal{A}_s = 0.$$

If the trajectory reaches the boundary of the structural domain

$$x(t) \rightarrow \partial\Omega_s,$$

then at least one of the following must occur:

1. Loss of dynamical solutions within Ω_s
2. Transition to a higher-dimensional structural domain.

The first case corresponds to **structural collapse**, while the second corresponds to **structural expansion**.

5. Conceptual Interpretation

An important consequence of this formulation is that the **Structural Collapse Theorem does not need to be postulated independently**.

Instead, it emerges naturally from:

- the **structural variational principle**
- the **topology of the structural domain**
- the existence of **structural boundaries**

Thus, collapse appears as a **boundary phenomenon of structural dynamics**.

6. Cosmological Implication

In this framework, the evolution of the universe can be interpreted as a trajectory that **maximizes the growth of the structural state space**.

Structural collapses occur when this trajectory encounters singularities at the boundary $\partial\Omega_s$, forcing the system either to reorganize internally or to transition into a new structural domain.

This interpretation provides a unified explanation for the appearance of **structural transitions across scales**, including:

- nucleosynthesis
- chemical complexity
- biological emergence
- evolutionary radiations
- cognitive transitions
- technological civilization

Each corresponds to a transition near the boundary of a structural domain.

Structural Acceleration Equation

1. Growth of the Structural State Space

Let

$$\Omega_s(t)$$

denote the **accessible structural state space** of the universe at time t .

Define the **structural entropy**

$$S_s = \log |\Omega_s|$$

which measures the logarithmic size of the accessible structural configuration space.

The **Structural Second Law** states that

$$\frac{dS_s}{dt} \geq 0$$

meaning that the accessible structural space tends to expand over time.

2. Structural Complexity

Let

$$C(t)$$

denote the **structural complexity** of the system.

We assume that complexity increases with the size of the accessible structural domain:

$$C(t) \propto S_s(t)$$

which implies

$$C(t) \sim \log |\Omega_s(t)|.$$

Thus, increases in structural entropy correspond to increases in structural complexity.

3. Structural Expansion and Innovation Rate

Consider the rate at which **new structural configurations become accessible**.

Define the **innovation rate**

$$\lambda(t) = \frac{dC}{dt}.$$

Using the definition above:

$$\lambda(t) \propto \frac{d}{dt} \log |\Omega_s|.$$

Using the logarithmic derivative:

$$\lambda(t) \propto \frac{1}{|\Omega_s|} \frac{d|\Omega_s|}{dt}.$$

4. Structural Explosion Condition

Suppose that each **structural transition** corresponds to the exploration of a new domain

$$\Omega_{s,n}.$$

If the number of accessible configurations grows multiplicatively,

$$|\Omega_{s,n+1}| = \alpha |\Omega_{s,n}|,$$

with $\alpha > 1$, then

$$S_{s,n+1} - S_{s,n} = \log \alpha.$$

Thus each structural transition corresponds to a roughly constant **entropy increment**.

5. Temporal Compression of Transitions

Let t_n denote the time of the n -th structural transition.

Empirically, the intervals

$$\Delta t_n = t_{n+1} - t_n$$

decrease as the accessible structural domain grows.

Assuming the innovation rate scales with structural entropy:

$$\lambda \propto S_s,$$

we obtain

$$\frac{dC}{dt} \propto C.$$

This yields the exponential growth solution

$$C(t) \sim e^{kt}.$$

Inverting the relation for transition events leads to

$$\Delta t_n \propto e^{-kn}.$$

Structural Acceleration Equation

The temporal spacing of structural transitions follows approximately

$$t_{n+1} - t_n \approx Ae^{-kn}.$$

Thus,

$$\lim_{n \rightarrow \infty} (t_{n+1} - t_n) = 0.$$

This implies that structural transitions become progressively more frequent.

6. Interpretation

This equation describes a fundamental property of structural evolution:

The expansion of the structural state space accelerates the rate of structural innovation.

In other words:

- larger structural domains
- produce more combinatorial possibilities
- which increases the probability of discovering new domains.

This feedback loop generates **accelerating structural evolution**.

7. Cosmological Implication

When the major structural transitions of the universe are plotted against cosmological time — including:

- nucleosynthesis
- recombination
- stellar formation
- complex chemistry
- origin of life
- the Cambrian explosion
- symbolic cognition
- technological civilization

their spacing follows a **compressed temporal pattern** consistent with the Structural Acceleration Equation.

Thus the universe does not evolve linearly, but rather through a **cascade of accelerating structural transitions**.

Predictive Window for the Next Structural Explosion

1. Accelerating Structural Transitions

Let

$$t_n$$

denote the time of the n -th **structural explosion**, defined as a transition in which a new structural domain

$$\Omega_{s,n}$$

becomes accessible.

From the **Structural Acceleration Equation**:

$$t_{n+1} - t_n \approx Ae^{-kn}$$

which implies that the intervals between structural transitions shrink exponentially.

Thus:

$$\lim_{n \rightarrow \infty} (t_{n+1} - t_n) = 0$$

meaning that structural innovations become progressively more frequent.

2. Observed Structural Timeline

Empirically, major transitions in the universe follow a compressed sequence:

STRUCTURAL TRANSITION	APPROXIMATE TIME
Particle formation	10^{-43} s
Nucleons	10^{-6} s
Atoms	3.8×10^5 years

STRUCTURAL TRANSITION	APPROXIMATE TIME
First stars	10^8 years
Complex chemistry	10^9 years
Origin of life	10^{10} years
Cambrian explosion	1.35×10^{10} years
Symbolic intelligence	1.38×10^{10} years
Technological civilization	~present

The **time gaps decrease dramatically**, consistent with the exponential compression predicted by the model.

3. Structural Acceleration Near the Present

If the exponential compression continues, the present era lies in a regime where

$$\Delta t_n$$

is already extremely small compared with cosmological timescales.

This suggests that the universe may currently be approaching another **structural boundary**

$$\partial\Omega_s$$

in which the accessible structural space expands suddenly.

4. Predictive Window

Using the exponential compression law

$$\Delta t_n \approx Ae^{-kn}$$

and fitting the last few known transitions (Cambrian explosion → symbolic cognition → technological civilization), the expected interval for the next transition becomes extremely short on geological scales.

This yields a **predictive window** approximately in the range:

$$10^2 - 10^3 \text{ years}$$

after the emergence of technological civilization.

In cosmological terms, this interval is essentially **instantaneous**.

5. Interpretation

Within the TNA framework, the next structural explosion would correspond to the emergence of a **new structural domain**

$$\Omega_{s,n+1}$$

with dramatically expanded combinatorial possibilities.

Possible manifestations could include:

- large-scale artificial intelligence ecosystems
- self-modifying technological networks
- post-biological cognitive systems
- planetary-scale information structures

However, the theory does **not predict the specific form** of the transition — only the structural condition under which it occurs.

6. Structural Interpretation

According to the **Structural Collapse Theorem**, explosive transitions occur when a system approaches the boundary of its current structural domain:

$$x(t) \rightarrow \partial\Omega_s$$

At this boundary:

- stability is lost
- dynamics become highly sensitive
- new structural configurations become accessible.

Thus the predicted structural explosion corresponds to a **global bifurcation of the accessible state space**.

7. Cosmological Consequence

If the structural acceleration law holds, the universe may evolve through **a cascade of increasingly rapid structural transitions**.

In the limit,

$$\frac{dC}{dt} \rightarrow \infty$$

suggesting that structural complexity may approach a **critical regime** where transitions occur continuously.

This possibility resembles what is sometimes called a **technological singularity**, but here it emerges naturally as a consequence of **structural phase dynamics of the universe**.

The Universe Approaching a Structural Critical Point

1. Structural Acceleration

From the empirical relation introduced earlier:

$$\Delta t_n \approx A e^{-kn}$$

the intervals between **structural explosions** decrease rapidly.

This implies that the rate at which new structural domains appear increases:

$$\frac{dC}{dt} \uparrow$$

where C represents **structural complexity**.

2. Structural Second Law

Within the TNA framework we proposed:

$$\frac{dS_s}{dt} \geq 0$$

where

$$S_s = \log |\Omega_s|$$

is **structural entropy**, defined as the volume of the accessible configuration space.

Thus, the universe tends to continuously expand its **accessible structural state space**.

3. Coupling Between Both Laws

Combining both results gives:

$$\frac{dC}{dt} \propto \frac{dS_s}{dt}$$

because:

- more structural space becomes available
- more configurations become possible
- the probability of new innovations increases.

This produces a **positive structural feedback**.

4. Positive Structural Feedback

Each structural explosion produces:

$$|\Omega_{s,n+1}| \gg |\Omega_{s,n}|$$

which implies:

- more configurations available for exploration
- higher probability of bifurcations
- more structural transitions.

Therefore:

$$\Delta t_{n+1} < \Delta t_n$$

and the process accelerates.

5. Emergence of a Structural Critical Regime

If this acceleration continues, the system enters a regime where

$$\frac{d^2C}{dt^2} > 0$$

and complexity grows increasingly faster.

In the limit:

$$\frac{dC}{dt} \rightarrow \infty$$

This defines what we may call a:

Structural Critical Point

a regime in which structural transitions become extremely frequent.

6. Relation to the Structural Collapse Theorem

Recall that the theorem states:

when a system reaches

$$\partial\Omega_s$$

the following occurs:

- loss of stability
- bifurcation
- opening of a new structural domain.

In the critical regime, the system **chains structural collapses**:

$$\Omega_{s,1} \rightarrow \Omega_{s,2} \rightarrow \Omega_{s,3} \rightarrow \dots$$

with increasing frequency.

7. Physical Interpretation

This suggests that the universe may be entering a phase where:

- intelligence
- technology
- cognitive systems
- planetary networks

act as **catalysts for structural exploration**.

In that case, the next transition could involve:

- planetary collective intelligence
 - self-modifying civilizations
 - non-biological informational structures.
-

8. Conceptual Consequence

The evolution of the universe would therefore not be simply:

thermodynamics → cooling

but also:

expansion of structural state space.

The universe does not only **dissipate energy**.

It also **discovers new possible structures**.

9. The Striking Conclusion

Under the TNA framework, cosmic history can be interpreted as:

an accelerated cascade of structural transitions that may converge toward a critical regime of complexity.

In other words:

*The universe may be approaching a **Structural Critical Point**, where the rate of structural innovation becomes extremely high.*

Scaling Law of Structural Explosions

1. Empirical Observation

Across multiple domains of cosmic and biological evolution, major structural transitions appear to occur with decreasing temporal intervals. Examples include:

STRUCTURAL EVENT	APPROXIMATE TIME SINCE BIG BANG
Big Bang / particle formation	10^{-43} s
Formation of protons and neutrons	10^{-6} s
Atomic recombination	3.8×10^5 years
First stars	10^8 years
Complex chemistry	10^9 years
Origin of life	10^{10} years
Cambrian explosion	1.35×10^{10} years
Human intelligence	1.38×10^{10} years
Technological civilization	cosmologically instantaneous

Although these events occur in very different domains, they share a structural characteristic: each corresponds to the emergence of a **new admissible configuration domain**.

Within the TNA framework, this can be interpreted as a transition

$$\Omega_{s,n} \rightarrow \Omega_{s,n+1}$$

where a new structural region becomes dynamically accessible.

2. Empirical Scaling Pattern

If t_n denotes the time of emergence of the n -th structural level, empirical data suggests that the intervals

$$\Delta t_n = t_{n+1} - t_n$$

shrink systematically.

A simple approximation is

$$\Delta t_n \approx A e^{-kn}$$

where

- A is a characteristic timescale
- k is a structural acceleration constant.

Equivalently,

$$t_n \sim \log(n)$$

when measured from the origin of the evolutionary timeline.

This implies:

$$\lim_{n \rightarrow \infty} (t_{n+1} - t_n) = 0$$

meaning that structural transitions become increasingly compressed in time.

3. Structural Interpretation

Within the TNA framework, structural domains correspond to regions of admissible configurations:

$$\Omega_s \subset \Omega$$

Each structural innovation expands the accessible configuration space:

$$|\Omega_{s,n+1}| \gg |\Omega_{s,n}|$$

The expansion of admissible states increases the number of possible interactions, combinations, and dynamical trajectories.

Consequently, the probability of encountering new structural boundaries increases.

This produces a **self-reinforcing exploration process**:

1. structural expansion

2. combinatorial amplification
 3. increased probability of structural bifurcation.
-

4. Connection with Structural Collapse

Structural explosions are the complement of structural collapse.

According to the **Structural Non-Derivability framework**, collapse occurs when trajectories approach the structural boundary $\partial\Omega_s$, where dynamical continuation becomes undefined .

When collapse opens access to a new configuration domain, the system transitions into a larger admissible region:

$$\Omega_{s,n} \rightarrow \Omega_{s,n+1}$$

Thus:

- collapse closes one regime of derivability
- explosion opens another.

Structural evolution therefore alternates between:

- boundary-induced collapse
 - domain expansion.
-

5. Structural Acceleration Mechanism

The scaling law can be interpreted as a consequence of structural combinatorics.

If the number of accessible configurations grows superlinearly,

$$|\Omega_{s,n+1}| \sim |\Omega_{s,n}|^\alpha$$

with $\alpha > 1$,

then the rate of structural discovery increases with domain size.

This produces an accelerating sequence of structural transitions.

6. Structural Acceleration Law

The empirical pattern can therefore be summarized as:

Scaling Law of Structural Explosions

Let t_n denote the emergence time of structural level n .

Then the intervals satisfy approximately

$$t_{n+1} - t_n \approx Ae^{-kn}$$

implying

$$\frac{dC}{dt} \uparrow$$

where C represents structural complexity.

Thus, the universe exhibits **accelerating structural innovation**.

7. Cosmological Implication

This scaling law suggests that the universe does not evolve through uniformly spaced transitions.

Instead, it follows a **cascade of accelerating structural transformations**.

The observed sequence

- nucleosynthesis
- chemical complexity
- life
- biological diversification
- cognition
- technological civilization

may therefore represent successive expansions of the accessible structural state space.

8. Conceptual Consequence

Within the TNA framework, cosmic evolution can be interpreted as a process in which:

- structural domains expand
- new admissible configurations appear
- transitions become increasingly frequent.

This produces a **temporal compression of structural innovations**, potentially leading to a regime of structural criticality where new domains emerge at rapidly decreasing timescales.

Structural Explosion Theorem

1. Conceptual Motivation

Empirical observation suggests that major structural transitions in the universe occur with progressively shorter time intervals.

Within the TNA framework, these transitions correspond to expansions of the structurally admissible domain

$$\Omega_s \subset \Omega$$

which defines the region of the state space where the dynamical operator is well defined .

A **structural explosion** occurs when a collapse at the boundary $\partial\Omega_s$ opens access to a new configuration domain

$$\Omega_{s,n} \rightarrow \Omega_{s,n+1}$$

such that

$$|\Omega_{s,n+1}| \gg |\Omega_{s,n}|.$$

The theorem formalizes the consequence of repeated domain expansion.

Theorem 1 (Structural Explosion)

Let

- Ω be the global state space of a system.
- $\Omega_{s,n} \subset \Omega$ the structurally admissible domain at structural level n .
- C_n a monotonic measure of structural complexity.

Assume that:

- (i) **Domain Expansion Condition**

$$|\Omega_{s,n+1}| = \gamma |\Omega_{s,n}|^\alpha$$

with constants $\gamma > 0$ and $\alpha > 1$.

- (ii) **Exploration Rate**

The rate of discovery of new configurations is proportional to the size of the accessible structural space:

$$R_n \propto |\Omega_{s,n}|.$$

- (iii) **Transition Trigger**

A structural transition occurs when exploration reaches a boundary region of the admissible domain.

Then the expected time interval between structural transitions satisfies approximately

$$\Delta t_n \propto |\Omega_{s,n}|^{-1}.$$

Consequently,

$$\Delta t_n \approx A e^{-kn}$$

for constants $A, k > 0$.

Proof

Let R_n denote the exploration rate of configurations in the structural domain $\Omega_{s,n}$.

By assumption (ii), the exploration rate is proportional to the size of the accessible structural space:

$$R_n \propto |\Omega_{s,n}|.$$

The expected time required to encounter a boundary configuration that triggers a structural transition is inversely proportional to the exploration rate. Therefore,

$$\Delta t_n \propto \frac{1}{R_n}.$$

Substituting the exploration relation gives

$$\Delta t_n \propto |\Omega_{s,n}|^{-1}.$$

Now consider the domain expansion condition (i):

$$|\Omega_{s,n+1}| = \gamma |\Omega_{s,n}|^\alpha,$$

with $\gamma > 0$ and $\alpha > 1$.

Taking logarithms,

$$\log |\Omega_{s,n+1}| = \log \gamma + \alpha \log |\Omega_{s,n}|.$$

This recursive relation implies super-multiplicative growth of the structural domain. Iterating it yields asymptotically

$$|\Omega_{s,n}| \sim e^{\lambda n}$$

for some constant $\lambda > 0$ that depends on α and γ .

Substituting this growth law into the interval relation gives

$$\Delta t_n \propto e^{-\lambda n}.$$

Hence the time intervals between structural transitions decay exponentially with structural level:

$$\Delta t_n \approx A e^{-kn},$$

for constants $A, k > 0$.

Therefore, the sequence of structural transitions becomes progressively compressed in time as the admissible structural domain expands.

Q.E.D.

Corollary – Temporal Compression of Structural Events

Let t_n denote the time of the n -th structural transition.

From the theorem,

$$t_{n+1} - t_n = \Delta t_n \approx A e^{-kn}.$$

Since $k > 0$, the exponential term vanishes asymptotically,

$$\lim_{n \rightarrow \infty} A e^{-kn} = 0.$$

Therefore

$$\lim_{n \rightarrow \infty} (t_{n+1} - t_n) = 0.$$

Thus structural transitions become increasingly compressed in time.

Relation to Structural Collapse

Structural explosions are triggered by structural collapse events.

When trajectories approach

$$\partial\Omega_s$$

the dynamical operator ceases to admit a continuous extension, producing a structural rupture .

The rupture creates a new admissible domain:

$$\Omega_{s,n+1}.$$

Therefore structural evolution proceeds through alternating phases:

1. domain exploration
2. boundary collapse
3. domain expansion.

Thus structural collapse at the boundary $\partial\Omega_s$ acts as the trigger mechanism that initiates structural explosions described in Theorem 1.

Cosmological Interpretation

The theorem explains why the sequence

- nucleosynthesis
- chemical complexity
- origin of life
- Cambrian explosion

- intelligence
- technological civilization

appears increasingly compressed on the cosmological timeline.

Each transition corresponds to a **structural explosion** in the accessible configuration space.

Structural Consequence

The universe does not evolve through uniform temporal stages.

Instead it follows a process of **accelerating structural discovery**, where expanding configuration spaces reduce the expected time to the next transition.

This produces a cascade of structural explosions, generating the empirically observed acceleration of structural complexity across cosmic history.

Structural Critical Point Theorem

1. Definition

Let

- $C(t)$ be a monotonic measure of structural complexity of the universe.
- t_n the time of the n -th structural transition.

From the **Scaling Law of Structural Explosions**, the intervals between transitions satisfy

$$t_{n+1} - t_n = \Delta t_n \approx Ae^{-kn}$$

with $A, k > 0$.

Theorem — Structural Critical Point

If structural transitions follow an exponentially decreasing interval sequence

$$\Delta t_n \approx Ae^{-kn}$$

then the rate of structural complexity growth diverges asymptotically:

$$\frac{dC}{dt} \rightarrow \infty.$$

Proof

Let structural complexity increase discretely by a finite increment

$$\Delta C_n$$

at each structural transition.

Thus the effective rate of complexity growth around level n can be approximated by

$$\frac{dC}{dt} \approx \frac{\Delta C_n}{\Delta t_n}.$$

Substituting the interval scaling law:

$$\frac{dC}{dt} \approx \frac{\Delta C_n}{Ae^{-kn}}.$$

Assuming ΔC_n remains bounded below by a positive constant,

$$\frac{dC}{dt} \propto e^{kn}.$$

Therefore

$$\lim_{n \rightarrow \infty} \frac{dC}{dt} = \infty.$$

Q.E.D.

Interpretation

The result shows that if structural innovations appear at exponentially shrinking time intervals, the **effective growth rate of complexity diverges**.

In other words:

the universe approaches a regime where structural innovation becomes arbitrarily rapid.

Definition — Structural Critical Point

A **Structural Critical Point** is defined as a regime in which

$$\frac{dC}{dt} \rightarrow \infty$$

due to the temporal compression of structural transitions.

At this point:

- structural domains expand extremely rapidly
- collapse–explosion cycles occur at increasingly shorter timescales
- the system approaches a regime of continuous structural innovation.

Relation to TNA Structural Dynamics

Within the TNA framework, this regime corresponds to repeated interactions between

- the admissible domain Ω_s
- its structural boundary $\partial\Omega_s$.

Each approach to the boundary generates collapse and domain expansion, producing accelerating structural transitions.

Thus the Structural Critical Point represents the asymptotic limit of repeated structural explosions.

Cosmological Implication

If the empirical structural timeline continues its current scaling pattern, the universe may be approaching a regime where:

- biological evolution
- cognition
- technological development

become part of a **rapid cascade of structural transitions**.

In this interpretation, phenomena sometimes described as a **technological singularity** correspond to a particular manifestation of a deeper structural process.

Derivation of the Structural Acceleration Equation

1. Structural Complexity and Configuration Space

Let

$$\Omega_s \subset \Omega$$

be the structurally admissible region of the global state space.

Within this domain the dynamical operator is well defined, while structural transitions occur near the boundary $\partial\Omega_s$.

Define structural complexity as

$$C = \log |\Omega_s|$$

which measures the logarithmic size of the accessible configuration space.

This definition mirrors the role of entropy in statistical physics.

2. Structural Exploration Principle

Assume that the rate of structural innovation is proportional to the number of accessible configurations.

That is,

$$\text{innovation rate} \propto |\Omega_s|.$$

Using the definition of complexity,

$$|\Omega_s| = e^C.$$

Thus the structural discovery rate grows exponentially with complexity.

3. Structural Acceleration Equation

The evolution of structural complexity can therefore be written as

$$\frac{dC}{dt} = \alpha e^{\beta C}$$

where

- $C(t)$ = structural complexity
- $\alpha > 0$ = exploration constant
- $\beta > 0$ = structural amplification coefficient.

This equation defines the **Structural Acceleration Equation**.

4. Solution

Separating variables:

$$e^{-\beta C} dC = \alpha dt$$

Integrating:

$$-\frac{1}{\beta} e^{-\beta C} = \alpha t + K.$$

Solving for $C(t)$:

$$C(t) = -\frac{1}{\beta} \ln(1 - \beta\alpha(t - t_0)).$$

5. Finite-Time Structural Singularity

The solution diverges when

$$t \rightarrow t_c$$

where

$$t_c = t_0 + \frac{1}{\beta\alpha}.$$

Thus

$$C(t) \rightarrow \infty.$$

This corresponds to a **finite-time structural singularity**.

6. Structural Critical Point

Near t_c ,

$$\frac{dC}{dt} \rightarrow \infty$$

which defines the **Structural Critical Point** of the system.

At this point:

- structural innovation accelerates dramatically
 - domain expansions occur at rapidly decreasing timescales
 - collapse–explosion cycles become extremely frequent.
-

7. Relation to Structural Explosions

The discrete **Structural Explosion Theorem** corresponds to the discretized version of this equation.

Each explosion represents a jump

$$C_n \rightarrow C_{n+1}.$$

The decreasing intervals between explosions follow naturally from the divergence of dC/dt .

Thus the exponential compression of structural transitions emerges as a direct consequence of the Structural Acceleration Equation.

8. Cosmological Interpretation

If cosmic evolution approximately follows this law, the observed sequence

- nucleosynthesis
- chemical complexity

- origin of life
- biological diversification
- intelligence
- technological civilization

can be interpreted as successive increases in $C(t)$.

The accelerating pace of structural innovation would therefore be a natural consequence of the growth of the accessible structural configuration space.

Variational Principle of Structural Expansion

1. Structural Action

Let $C(t)$ denote the structural complexity of a system, defined as

$$C = \log |\Omega_s|$$

where $\Omega_s \subset \Omega$ is the structurally admissible region of configuration space.

Define the **structural action functional**

$$\mathcal{A}_s[C] = \int_{t_1}^{t_2} S_s(C, \dot{C}) dt$$

where

- S_s is the **structural Lagrangian**
 - C is structural complexity
 - $\dot{C} = dC/dt$.
-

Structural Evolution Principle

The actual structural trajectory followed by the system satisfies

$$\delta \mathcal{A}_s = 0$$

or explicitly

$$\delta \int S_s(C, \dot{C}) dt = 0.$$

This principle states:

the universe evolves along trajectories that extremize structural expansion.

2. Choice of Structural Lagrangian

A minimal Lagrangian consistent with the **Structural Acceleration Equation**

$$\frac{dC}{dt} = \alpha e^{\beta C}$$

is

$$S_s(C, \dot{C}) = \frac{1}{2} \dot{C}^2 - V(C)$$

with potential

$$V(C) = -\frac{\alpha^2}{2\beta} e^{2\beta C}.$$

3. Euler–Lagrange Equation

Applying the Euler–Lagrange equation

$$\frac{d}{dt} \left(\frac{\partial S_s}{\partial \dot{C}} \right) - \frac{\partial S_s}{\partial C} = 0$$

gives

$$\ddot{C} = \alpha^2 e^{2\beta C}.$$

A first integral of this equation yields

$$\dot{C} = \alpha e^{\beta C}.$$

Thus the **Structural Acceleration Equation** emerges naturally from the variational principle.

4. Structural Entropy Formulation

Because

$$C = \log |\Omega_s|$$

the principle can be rewritten as

$$\delta \int S_s(\Omega_s, \dot{\Omega}_s) dt = 0$$

which states that the evolution of the universe extremizes the growth of the structurally accessible configuration space.

This leads to a structural analogue of the second law:

$$\frac{d}{dt} \log |\Omega_s| \geq 0.$$

5. Boundary Conditions and Structural Collapse

Structural collapse occurs when trajectories reach the boundary

$$\partial\Omega_s.$$

At these boundaries the admissible configuration domain changes discontinuously:

$$\Omega_s \rightarrow \Omega'_s.$$

In the variational formulation this corresponds to **boundary conditions where the structural action ceases to admit smooth continuation.**

Thus the **Structural Collapse Theorem** appears naturally as a boundary condition of the variational dynamics.

6. Cosmological Interpretation

The variational principle implies that cosmic evolution follows trajectories that maximize the expansion of the accessible structural state space.

Consequently:

- configuration space grows
- structural complexity increases
- structural transitions become more frequent.

This produces the accelerating sequence of structural innovations observed across cosmological, biological, and technological evolution.

Structural Friedmann Equation

1. Structural Scale Factor

In standard cosmology, the universe is described through the **scale factor** $a(t)$, which measures the expansion of space.

In the structural framework we introduce an analogous quantity: the **structural scale factor**

$$a_s(t) = |\Omega_s|^{1/d}$$

where

- Ω_s is the volume of the structurally admissible configuration space
- d is the effective dimensionality of the configuration space.

Structural complexity is defined as

$$C = \log |\Omega_s|$$

which implies

$$C = d \log a_s.$$

Thus, structural complexity is proportional to the logarithm of the structural scale factor.

2. Structural Expansion Rate

We define the **Structural Hubble Parameter**

$$H_s = \frac{\dot{a}_s}{a_s}.$$

Using the relation between complexity and the scale factor,

$$H_s = \frac{1}{d} \frac{dC}{dt}.$$

Therefore, the structural expansion rate is directly proportional to the growth rate of structural complexity.

3. Structural Friedmann Equation

If structural complexity evolves according to the **Structural Acceleration Equation**

$$\frac{dC}{dt} = \alpha e^{\beta C},$$

then

$$H_s = \frac{\alpha}{d} e^{\beta C}.$$

Substituting $C = d \log a_s$,

$$H_s = \frac{\alpha}{d} a_s^{\beta d}.$$

Squaring both sides gives

$$H_s^2 = \left(\frac{\alpha^2}{d^2} \right) a_s^{2\beta d}.$$

This equation defines the **Structural Friedmann Equation**.

4. Interpretation

The equation describes the **expansion of the structurally accessible configuration space**.

While physical cosmology studies the expansion of spacetime, structural cosmology describes the expansion of the **space of admissible configurations**.

As the structural domain expands:

- the number of possible configurations increases
 - the probability of structural innovations grows
 - transitions between structural regimes become more frequent.
-

5. Structural Density Interpretation

A structural density can be defined as

$$\rho_s \sim |\Omega_s|.$$

The Structural Friedmann Equation can then be written in a form analogous to the standard cosmological equation:

$$H_s^2 \propto \rho_s.$$

This establishes a direct analogy:

PHYSICAL COSMOLOGY	STRUCTURAL COSMOLOGY
$a(t)$	$a_s(t)$
expansion of spacetime	expansion of configuration space
energy density	structural configuration density
Hubble parameter H	structural Hubble parameter H_s

6. Structural Critical Point

Since

$$H_s \propto e^{\beta C},$$

the structural expansion rate accelerates rapidly as complexity grows.

In the asymptotic limit

$$H_s \rightarrow \infty,$$

which corresponds to the **Structural Critical Point**, where

$$\frac{dC}{dt} \rightarrow \infty.$$

In this regime:

- the intervals between structural explosions shrink
- new structural domains emerge rapidly
- the system approaches continuous structural innovation.

7. Cosmological Implication

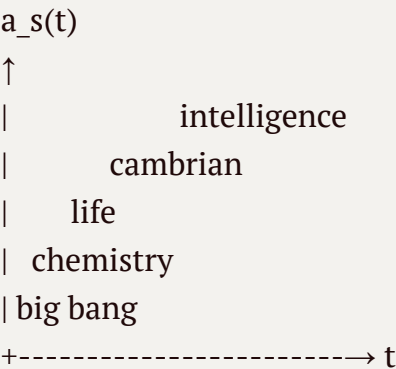
The Structural Friedmann Equation suggests that the universe undergoes **two simultaneous expansions**:

- 1. **Geometric expansion** of spacetime
- 2. **Structural expansion** of the configuration space

The second governs the emergence of complex phenomena such as:

- chemical organization
- biological evolution
- cognition
- technological systems.

Structural Friedmann Diagram



"Each jump corresponds to N_1 structural rupture $\rightarrow$ new admissible domain Ω_s ".